

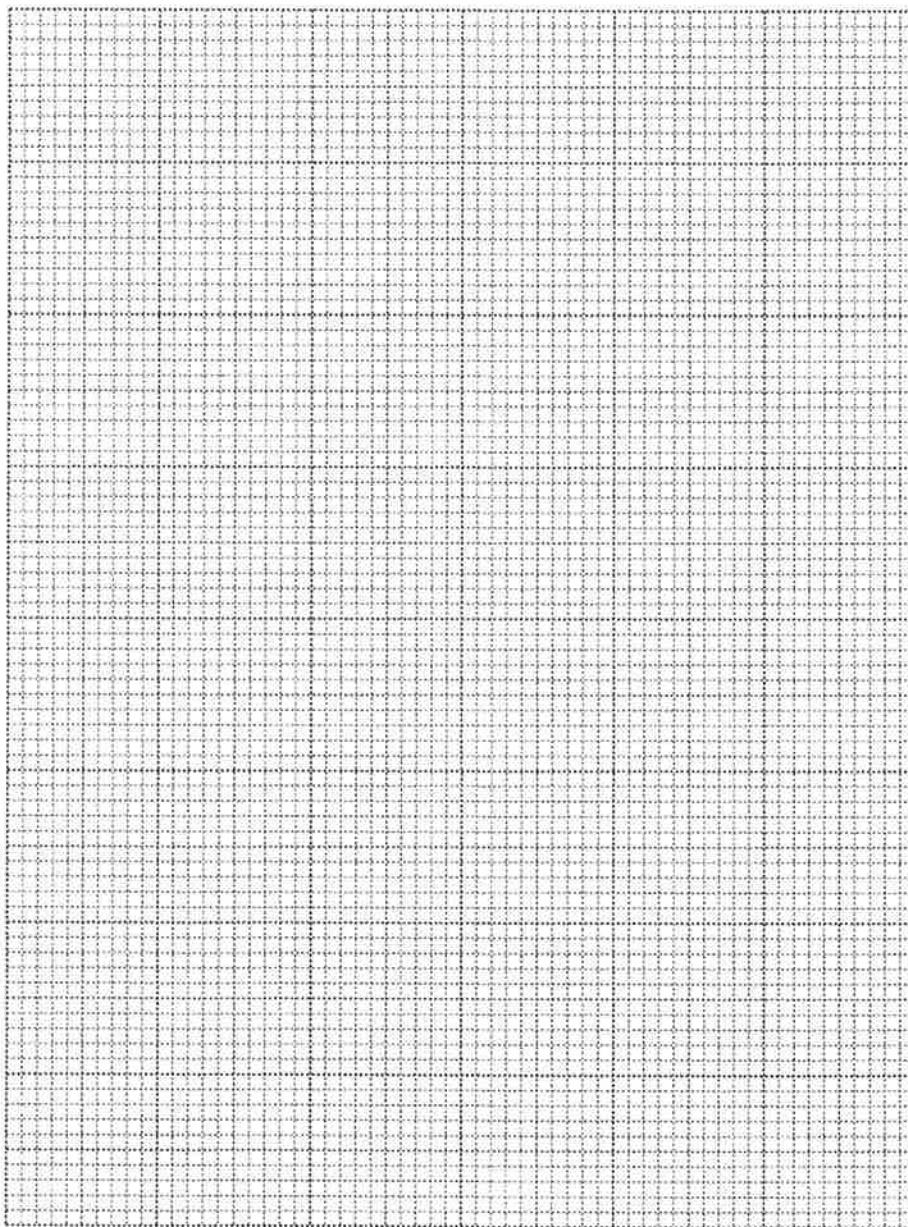
Intermediate

1. When a substance, B , is dissolved in acid, a reaction occurs. The amount of B , m grams, present at time t minutes after the start of the reaction is believed to be given by the formula $m = Ae^{-kt}$, where A and k are constants. Measurements of m and t are shown in the table below.

t	1	2	3	4	5
m	15.6	12.1	9.5	7.4	5.7

- (a) Plot $\ln m$ against t and draw a straight line graph to illustrate the information.

[2]



- (b) Use your graph to estimate the amount of B present at the start of the reaction.
- (c) Use your graph to estimate the time taken, to the nearest second, when 50% of B has been dissolved.

[2]

[2]

[N22/I/2]

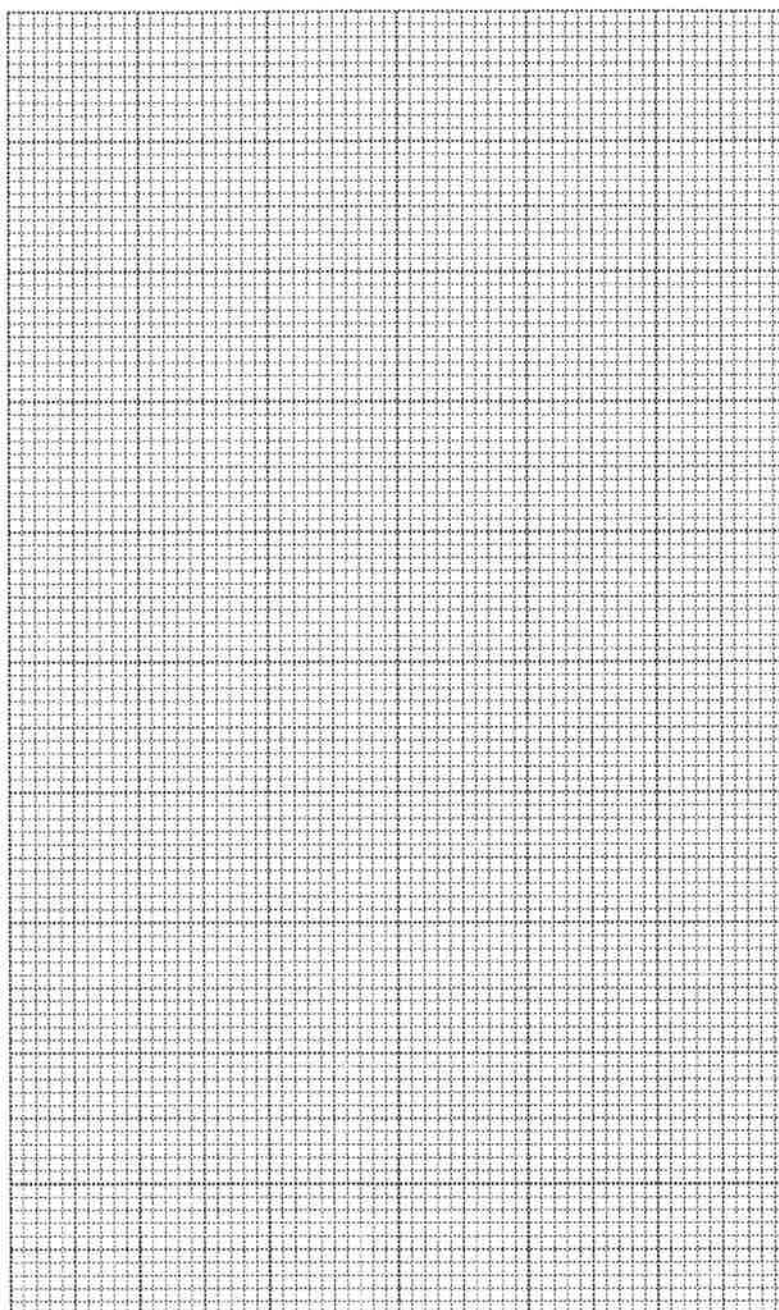
2. (a) A formula for working out the stopping distance, d , for a vehicle travelling at a speed v , is $d = av^2 + bv$, where a and b are constants. Values of d for different values of v have been collected. Explain how a straight line graph can be drawn to represent the formula, and state how the values of a and b could be obtained from the line. [4]
- (b) Since 1960, the tiger population in an Asian country has been steadily decreasing. The table shows the estimated number of tigers, n , in the decades following 1960.

The decade 1960–1969 is taken as $t = 1$, and so on.

Year	1960–1969	1970–1979	1980–1989	1990–1999
Value of t	1	2	3	4
Number of tigers n	810	450	240	135

A wild-life expert believed that these figures can be modelled by the formula $n = ab^t$, where a and b are constants.

- (i) Draw a straight line graph to show that the model is reasonable. [4]



- (ii) Estimate the number of tigers in the decade 2000–2009. [2]
- (iii) Give a reason why this model might not be accurate in later decades. [1]

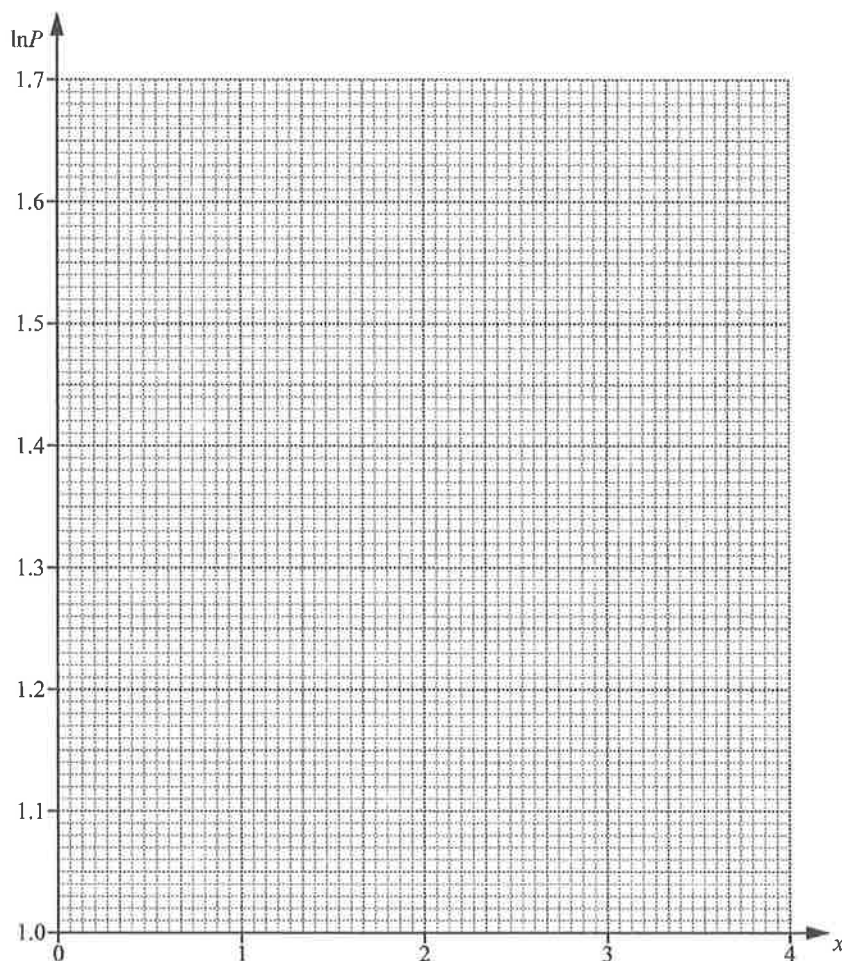
[N21/II/7]

3. The table shows, to 3 significant figures, the population, P , in millions, of a country on January 1st at intervals of five years from 1995 to 2015. The variable x is measured in units of 5 years.

Year	1995	2000	2005	2010	2015
x	0	1	2	3	4
P	3.32	3.75	4.24	4.80	5.43

- (i) On the grid below plot $\ln P$ against x and draw a straight line graph.

[2]



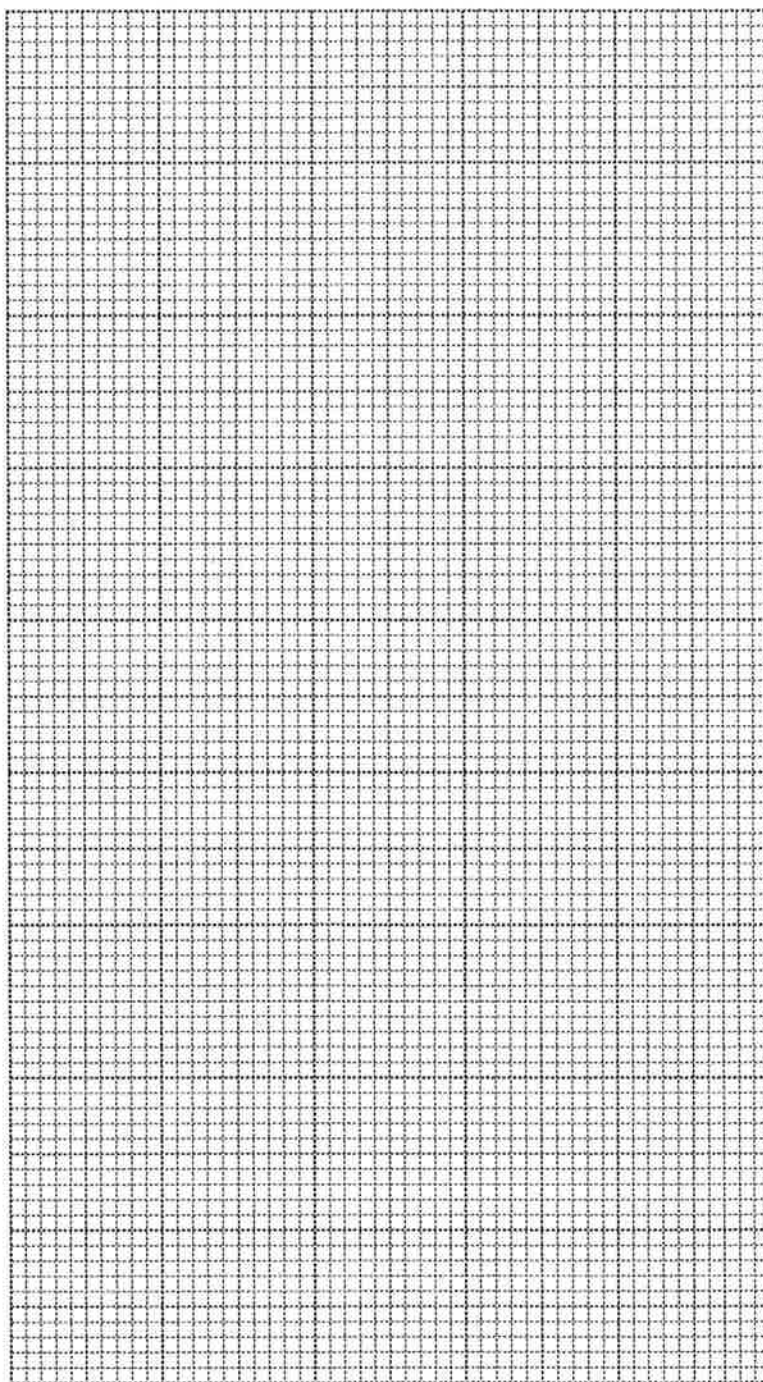
- (ii) Find the gradient of your straight line and hence express P in the form Ae^{kx} , where A and k are constants. [4]
- (iii) If this model for the population remains valid, find the first year of the interval in which the population first exceeds 8 million. [3]

[N20/II/6]

4. The speed, v m/s, of a vehicle, t s after passing a fixed point O , is given, for $t \geq 0$, by $v = Ae^{\frac{t}{k}} - 3$, where A and k are constants. The table below shows corresponding values of t and v .

t	2	4	6	8
v	6.35	4.38	2.67	1.41

- (i) Draw the graph of $\ln(v + 3)$ plotted against t , using a scale of 1 cm for 1 unit on the t -axis and a scale of 5 cm to 1 unit on the $\ln(v + 3)$ -axis. [3]



- (ii) Use the graph to estimate the value of each of the constants A and k . [5]
 (iii) State the speed of the vehicle at O . [1]
 (iv) Explain how the graph could be used to find the value of t when the speed of the vehicle is 5 m/s. [1]
 (v) Using your values of A and k , calculate the value of t when the vehicle comes to rest. [1]

[N19/II/8]

5. Recorded values of the mass, m grams, of a radioactive substance, t hours after observations began, are shown in the table below.

m (grams)	36.4	22.1	13.4	8.1
t (hours)	20	40	60	80

It is known that m and t are related by the equation $m = m_0 e^{-kt}$, where m_0 and k are constants.

- (i) Plot $\ln m$ against t and draw a straight line graph. [3]

Use your graph to estimate

- (ii) the mass of the substance when the observations began, [2]
 (iii) the value of k , [2]
 (iv) the time taken for the substance to lose half of its original mass. [2]

[N18/II/4]

6. The variables x and y are such that when values of $\frac{1}{y}$ are plotted against $\frac{1}{\sqrt{x}}$ a straight line is obtained. It is given that $y = 0.25$ when $x = 0.04$ and that $y = 0.50$ when $x = 1.00$. Find the value of y when $x = 9$. [5]

[N17/I/3]

7. The price, $\$P$, of a company share on 1st January has been increasing each year from 1995 to 2015. The company claims that this increase is exponential and so can be modelled by an equation of the form

$$P = P_0 e^{kt},$$

where P_0 and k are constants and t is the time in years since 1st January 1995. The table below gives values of P and t for some of the years 1995 to 2010.

Year	1995	2000	2005	2010
t years	0	5	10	15
$\$P$	2.00	2.44	3.00	3.65

- (i) Plot a suitable straight line graph to show that the model is valid for the years 1995 to 2010. [2]
 (ii) Estimate the value of P_0 and k . [3]
 (iii) Assuming that the model is still appropriate, estimate the price of a share on 1st January 2015. [2]

[N16/II/1]

Advanced

8. A cuboid of volume $V \text{ cm}^3$ has a height of $x \text{ cm}$ and a rectangular base of area $(px^2 + q) \text{ cm}^2$. Corresponding values of x and V are shown in the table below.

x	5	10	15	20
V	175	650	1725	3700

- (i) Using suitable variables, draw, on graph paper, a straight line graph and hence estimate the value of each of the constants p and q . [6]
- (ii) Using your values of p and q , calculate the value of x for which the cuboid is a cube. [2]
- (iii) Explain how another straight line drawn on your diagram can lead to an estimate of the value of x for which the cuboid is a cube. Draw this line and hence verify your value of x found in part (ii). [3]

[N15/II/11]